School Worksheet

Class 12 Mathematics, Chapter 5: Continuity and Differentiability

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

The value of k for which $f(x) = \begin{cases} kx + 1, & x \le 5 \\ 3x - 5, & x > 5 \end{cases}$ is continuous at $x = 5$ is:

(A) $\frac{9}{5}$

(B) $\frac{5}{9}$

(C) $\frac{11}{5}$

(D) 2

2 MCQ — 1 mark

The function $f(x) = |x|$ is:

(A) continuous and differentiable everywhere

(B) continuous everywhere but not differentiable at $x = 0$

(C) differentiable but not continuous at $x = 0$

(D) neither continuous nor differentiable at $x = 0$

3 MCQ — 1 mark

The derivative of e^{e^x} with respect to x is:

(A) $\log x$ **(B)** e^{e^x}

(C) $e^x \cdot e^{e^x}$ **(D)** $(e^x)^2$

4 MCQ — 1 mark

If $y = \log(\cos e^x)$, then $\frac{dy}{dx}$ is:

(A) $\tan e^x$ **(B)** $-e^x \tan e^x$

(C) $e^x \tan e^x$ **(D)** $-\tan e^x$

Q No.	Question	Marks
5	MCQ If $x = at^2$ and $y = 2at$, then $\frac{dy}{dx}$ is: **(A)** t **(B)** $\frac{1}{t}$ **(C)** $2t$ **(D)** $-\frac{1}{t^2}$	1 mark
6	Assertion-Reason **Assertion (A):** The greatest integer function $f(x) = [x]$ is discontinuous at $x = 2$. **Reason (R):** $\lim_{x \to 2^-} [x] \neq \lim_{x \to 2^+} [x]$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer Differentiate $\sin^{-1}\left(2x\sqrt{1-x^2}\right)$ with respect to x, where $-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$.	2 marks
Level 2		**18 marks**
8	Short answer Find k so that $f(x) = \begin{cases} \frac{1-\cos 4x}{8x^2}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$.	2 marks
9	Short answer Find $\frac{dy}{dx}$ if $y = x^{\sin x} + (\sin x)^x$.	3 marks
10	Short answer If $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$, find $\frac{dy}{dx}$.	2 marks
11	Short answer If $y = e^{a\cos^{-1}x}$, $-1 < x < 1$, show that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0$.	3 marks
12	Short answer If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, show that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.	3 marks
13	Short answer Find $\frac{dy}{dx}$ if $x^2 + 2xy + y^3 = 42$.	2 marks

Q No.	Question	Marks
14	Short answer Find the values of a and b such that $f(x) = \begin{cases} 5, & x \le 2 \\ ax + b, & 2 < x < 10 \\ 21, & x \ge 10 \end{cases}$ is continuous.	3 marks

Level 3

25 marks

15	Short answer If $y = (\tan^{-1} x)^2$, show that $(x^2+1)^2 \dfrac{d^2y}{dx^2} + 2x(x^2+1)\dfrac{dy}{dx} = 2$.	3 marks
16	Long answer If $(x-a)^2 + (y-b)^2 = c^2$ for some $c > 0$, prove that $\dfrac{\left[1 + \left(\dfrac{dy}{dx}\right)^2\right]^{3/2}}{\dfrac{d^2y}{dx^2}}$ is a constant independent of a and b.	5 marks
17	Short answer If $x^y = e^{x-y}$, prove that $\dfrac{dy}{dx} = \dfrac{\log x}{(1+\log x)^2}$.	3 marks
18	Long answer Show that $f(x) = \lvert x-1\rvert + \lvert x-2\rvert$ is continuous at $x = 1$ and $x = 2$ but not differentiable at these points.	5 marks
19	Long answer If $x = a\left(\cos\theta + \log\tan\dfrac{\theta}{2}\right)$ and $y = \sin\theta$, then find $\dfrac{d^2y}{dx^2}$ at $\theta = \dfrac{\pi}{4}$.	5 marks
20	Case study **Case Study:** While designing a smooth ramp, an engineer models its profile by $f(x) = \begin{cases} x^2 + 3x + p, & x \le 1 \\ qx + 2, & x > 1 \end{cases}$. For a smooth ramp, f must be differentiable at $x = 1$. **(i)** Write the condition on p and q for f to be continuous at $x = 1$. — 1 mark **(ii)** Find the left-hand derivative and right-hand derivative of f at $x = 1$. — 1 mark **(iii)** Find the values of p and q. — 2 marks	4 marks

Answer key

Chapter 5: Continuity and Differentiability, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) $\frac{9}{5}$
2	(B) continuous everywhere but not differentiable at $x = 0$
3	(C) $e^x \cdot e^{e^x}$
4	(B) $-e^x \tan e^x$
5	(B) $\frac{1}{t}$
6	(A) Both A and R are true and R is the correct explanation of A
7	$\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$

Level 2

8	$k = 1$
9	$\frac{dy}{dx} = x^{\sin x}\left(\cos x \log x + \frac{\sin x}{x}\right) + (\sin x)^x \left(\log \sin x + x \cot x\right)$
10	$\frac{dy}{dx} = \tan \frac{\theta}{2}$
11	$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2 y = 0$ (proved)
12	$\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$ (proved)
13	$\frac{dy}{dx} = -\frac{2(x+y)}{2x+3y^2}$
14	$a = 2, b = 1$

Level 3

15	$(x^2+1)^2\frac{d^2y}{dx^2} + 2x(x^2+1)\frac{dy}{dx} = 2$ (proved)
16	The expression equals $-c$ (or c when $y < b$), a constant independent of a and b.
17	$\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$ (proved)
18	f is continuous at $x = 1$ and $x = 2$; at $x = 1$, LHD $= -2 \neq 0 =$ RHD; at $x = 2$, LHD $= 0 \neq 2 =$ RHD, so not differentiable there.
19	$\frac{d^2y}{dx^2} = \frac{2\sqrt{2}}{a^2}$ at $\theta = \frac{\pi}{4}$
20	(i) $4 + p = q + 2$; (ii) LHD $= 5$, RHD $= q$; (iii) $p = 3$, $q = 5$